While providing the prior ranges, we expect them to contain error budgets from waveform systematics, missing physics, and other data analysis artifacts. Examples of the data analysis artifacts include residual power from glitch removal near the signal or non-stationary noise. We can integrate other approaches in the literature with our framework to provide error budgets for waveform systematics, thus serving as a complementary method. Another unique feature of our framework is the ability to mitigate type-A systematic errors. If these types of systematic errors are present, they can show up in the posterior distribution of the nodal points as deviating away from $(\delta\tilde{A}_i, \delta\tilde{\phi}_i) = 0$. It can guide determining that type-A systematic errors are present in the data, as shown in panel (c) of Figure 8. For type-B errors mitigation, we can combine our framework with the methods that account for differences between waveform models.

VII. DISCUSSION AND SUMMARY

The systematic errors in the PE analysis of a GW merger can arise due to waveform modeling errors, data analysis artifacts, or missing physics. As the GW detectors become more sensitive, the statistical errors become smaller, and we are reaching an era where systematic errors can not be ignored. Several approaches in the literature deal with accounting potential systematic errors in PE. To account for potential differences between existing waveform models, posterior samples are combined [6, 33, 34], or hyperparameter models are used to sample likelihood samples [35–38, 40]. Other approaches include marginalizing the potential differences using prior distribution analytically [41]. These priors can be constructed by estimating the NR calibration errors [42, 43] or using PN-based methods to mitigate the errors [45].

In this work, we present a general PE framework to account for the errors in the waveform modeling of GW sources. We assume that the reference signal, $h_{\rm ref}$, used in the recovery of the GW source parameters, is different from the 'true' signal $h_{\rm true}$; the parameters can model the differences $(\delta\tilde{A}, \delta\tilde{\phi})$, as differences in the amplitude and the phase of the signal. We present two WF-Error parametrizations: one corresponds to the relative phase, while the other corresponds to the absolute phase difference from the so-called true phase.

We use the cubic-spline method to modify the reference waveform model. We developed a Python code as a plugin that can be easily integrated with the publicly available GW analysis tool PyCBC. The source code can be downloaded from [114]. We also release the frame files for the simulated injections and configuration files of all the PE runs done in this analysis [123].

If the WF-Error budgets for the parameters $(\delta\tilde{A}, \delta\tilde{\phi})$ are available for a given waveform model, these error budgets can be used as a prior in WF-Error parametrization. Without such error budgets, we can use wider priors, potentially making our errors in estimating other parameters broader. This code and analysis framework can be utilized for the following purposes: i) to account for systematic errors in waveform models by incorporating appropriate priors for $(\delta\tilde{A}, \delta\tilde{\phi})$, such as calibration errors from numerical relativity (NR), or ii) to address the potential bias that may arise from missing physical effects in the waveform description.

We use Fisher matrix formalism to study its abilities and limitations in accounting for the bias induced by $(\delta\tilde{A}, \delta\tilde{\phi})$. In the LSA regime, where the modified model is very close to the reference model, we can use the Fisher matrix formalism to account for systematic biases. However, in the non-LSA regime, where the overlap between the waveform models is smaller, we can no longer trust the Fisher matrix approach to correctly predict the systematic bias.

Our findings indicate that even a one-percent phase error in the rel-phase parametrization can introduce bias in the observed chirp mass. The bias in the chirp mass could affect the conclusion of follow-up studies that rely on the correct distribution of source parameters, such as inferring the properties of an intrinsic population of BBH mergers. For the abs-phase parametrization, the moderate deviation in $\delta\phi$ of the order of $\sim \mathcal{O}(0.01)$ radians, with cubic splines, do not introduce significant bias, at least in current generation detectors. We need a more significant deviation or a very high SNR signal to notice any bias. However, even in the absence of significant bias, we observe a broadening of the posterior samples by up to 20 percent ($\frac{\sigma}{\sigma_{\rm ref}} \approx 0.2$), where σ is the standard deviation of 1D marginalized posterior of parameter $\mathcal{M}$ using the abs-phase parametrization and $\sigma_{\rm ref}$ is the standard deviation with the reference waveform model.

We make a case for waveform developers to provide error budgets alongside the waveform models. It is necessary to accurately account for the WF-Error parametrization in the PE analysis. These error budgets can be frequency dependent $(\delta\tilde{A}, \delta\tilde{\phi})$ regions of 1σ uncertainties. We call them WF-Error envelopes. These WF-Error envelopes are expected to be a function of parameter space. These envelopes can be used as priors for $(\delta\tilde{A}, \delta\tilde{\phi})$ parameters in WF-Error parametrization. However, if such envelopes are unavailable, we can use wider priors to account for potential waveform systematics, allowing the signal's loudness and data to inform us about the constraints on the WF-Error parameters. In this scenario, we may face a penalty in the form of broader posterior samples due to the significantly larger prior volume.

One of the use cases of WF-Error parametrization is to see if the waveform model in use is missing a description of reality. The next generation of waveform models are expected to include combined effects such as eccentric orbits, precessing systems, and higher-order modes. If any of the above-mentioned effects are missing from the description of the waveform model, we might introduce bias in inferred parameters such as chirp mass. Assuming the true model differs from the reference waveform model

used for the PE, we can use WF-Error parametrization to determine if the data favors this hypothesis. At the very least, we expect the posterior samples to be broad enough to bring down the bias and standard deviation ratio. In order to test this, we used ten random realizations of mildly processing systems and performed PE analysis with the non-spinning model. We find expected bias in this scenario. However, when we switch on rel-phase parametrization, the ratio $|\Delta\mathcal{M}^{\rm z}|/\sigma$ comes down, up to, by a factor of three. In this case, the abs-phase parametrization could not account for correcting the bias with given prior range.

The method we present here is based on certain assumptions that need to be tested in broader scenarios. We assume that the reference waveform differs from the true waveform in a manner that can be parameterized using one of the WF-Error parametric models described by equations (12) or (13). Additionally, we assume that the number of knots or nodal points selected within the frequency bins are sufficient to produce cubic spline curves that accurately model the deviation.

As we approach an era where loud signals are more common, we expect systematic biases to become amplified. It is important to account for the WF-Error parametrization or any other scheme that can address potential systematics in the reference waveform model. The pipeline we present can work in scenarios where the types of systematics are unknown. We can select our preferred waveform model, and the parametrization best describes the deviations, allowing the data to indicate any possible systematic errors. At the same time, this approach results in broader posterior samples, and such an outcome is anticipated.

The authors plan to explore the use of WF-Error parametrization to correct for potential biases in source parameters due to missing physics in more detail in the follow-up studies. For example, the next generation waveform models are expected to include the effects such as spin-precession, eccentric orbits, as well as higher modes. Until this is achieved, there might always be a question of whether there is inherent bias in estimating source properties due to a missing description of one of the above effects. In our follow-up studies, we will use existing NR simulation and state-of-the-art waveform models to study if any potential biases can be accounted for, and corrected if possible.

We make certain assumptions for the method presented here, and these assumptions need to be tested in broader scenarios. We assume that the reference waveform differs from the true waveform in a manner that can be parameterized using one of the WF-Error parametric models described by equations (12) or (13). Additionally, we assume that the number of knots or nodal points selected within the frequency bins are sufficient to produce cubic spline curves that accurately model the deviation. In our future studies, we will thoroughly test these assumptions for more general scenarios such as the systems with precession and/or eccentricity. Moreover, if the amplitude and phase corrections are highly oscillating functions of frequency, which can not be captured by the cubic-spline curves and a small number of fixed nodal points, we need to test other schemes, such as flexible nodal points and/or more complicated interpolation schemes.

ACKNOWLEDGMENTS

We acknowledge the Max Planck Gesellschaft and the Max Planck Independent Research Group Program, through which this work was supported. SK also acknowledges support by the research programme of the Netherlands Organisation for Scientific Research (NWO). We thank the computing team from AEI Hannover for their significant technical support. We also thank Binary Merger Observations and NR group members at AEI for their feedback and valuable comments. We further thank Francesco Jimenez Forteza for in depth discussions and feedback. We are also grateful to Krishnendu NV, Prayush Kumar, and Chris Van Den Broeck for discussions and valuable inputs. We thank Michael Pürrer for valuable comments on an earlier version of this manuscript. We thank the anonymous referee for their valuable feedback. We used the Holodeck computing cluster and the Atlas cluster at AEI Hannover for all the computation.

Appendix A: Fisher Matrix Formalism

In section IV, we have applied the Fisher matrix formalism to quantify systematic biases, but we have assumed a familiarity with the topic. This section serves as an introduction to the formalism itself.

The framework in which Fisher matrix estimates are derived is based on a geometric perspective on PE. Using a metric on the parameter space, the Fisher matrix

$$\Gamma_{\mu\nu} = \langle \partial_\mu h, \partial_\nu h \rangle \,, \tag{A1}$$

one can derive estimates about various errors in the process.[1] Here, h is some fiducial waveform model and ∂_μ refers to the derivative with respect to the parameter Θ^μ. Eq. (A1) is defined in terms of the noise-weighted inner product

$$\langle a, b \rangle = 4\Re \int_{f_{\rm low}}^{f_{\rm high}} \frac{\tilde{a}(f)\,\tilde{b}^*(f)}{S_n(f)}\, df \,, \tag{A2}$$

which has already been used in Eq. (3). It is this inner product that several important notions in GW data

[1] As far as we can see, Ref. [124] was the first one to use these estimates. The related geometric interpretation was first applied to GW data analysis by Ref. [125].

analysis are based on. Specifically, we will use 'match' to refer to the inner product optimized over relative time and phase shifts t_0, ϕ_0 between a, b and 'overlap' to refer to the normalized match, where the result is divided by the product $||a|| \cdot ||b||$. A complementary notion is the 'mismatch', which is defined as $1 -$ match.

The Fisher matrix formalism is a very popular tool because it allows to estimate biases in the source parameters Θ, which we will denote as $\Delta\Theta = \Theta_{\rm bf} - \Theta_{\rm tr}$. The two parameters $\Theta_{\rm tr}, \Theta_{\rm bf}$ are to be understood in the following context: say we have data that contains a signal produced by a model h_1 (and potentially some noise on top of that), and we wish to run PE on this data using another model h_2; then we call the parameters that h_1 is evaluated in the 'true parameters' $\Theta_{\rm tr}$ and the parameters obtained from PE 'best-fitting' parameters $\Theta_{\rm bf}$. What the Fisher matrix formalism does for us is that it allows to obtain an estimate of the parameter difference between true and recovered parameters without having to perform a full PE run. This difference will have two contributions: the systematic bias due to $h_1 \neq h_2$ can be estimated as [124]

$$\Delta\Theta^{\mu}_{\rm sys} = \sum_{\nu} (\Gamma^{-1})^{\mu\nu} \langle h_1 - h_2, \partial_\nu h_2 \rangle \tag{A3}$$

and the corresponding measurement uncertainty caused by noise (which is also called statistical bias) as [124]

$$\Delta\Theta^{\mu}_{\rm stat} = \sum_{\nu} (\Gamma^{-1})^{\mu\nu} \langle n, \partial_\nu h_2 \rangle \,. \tag{A4}$$

In reality, where n is not known, it is more common to work with the corresponding standard deviation [124]

$$\sigma_{\Delta\Theta^{\mu}_{\rm stat}} = \sqrt{(\Gamma^{-1})^{\mu\mu}} \,. \tag{A5}$$

These estimates are very valuable because they can be calculated without having to do an extra PE run. The right hand side of both equations can be evaluated either in the true, injected parameters $\Theta_{\rm tr}$ or in the best-fitting parameters $\Theta_{\rm bf}$ that would be the result of a PE run using h_2 on the signal $h_1(\Theta_{\rm tr})$. To calculate the Fisher matrix in this context, h_2 must be used.

However, there is also a caveat. The whole approach depends crucially on the validity of the linear signal approximation (LSA)

$$h_2(\Theta_{\rm tr}) \simeq h_2(\Theta_{\rm bf}) + \partial_\mu h_2(\Theta_{\rm bf})(\Theta_{\rm tr} - \Theta_{\rm bf})^{\mu} \,. \tag{A6}$$

Besides numerical issues with condition numbers of the Fisher matrix, which are also common problems, this is the major bottleneck of Fisher matrix estimates. If h_1 and h_2 exhibit large differences, they might produce maximum-a-posteriori estimates $\Theta_{\rm tr}, \Theta_{\rm bf}$ for which (A6) is not a good approximation. In this case, the accuracy of later applied estimates such as (A3) may be compromised.

Recently, the authors of [28] introduced a scheme to improve this issue, which we have adopted in this paper as well. The idea, which we only briefly review, is to replace the waveform difference $h_1(\Theta_{\rm tr}) - h_2(\Theta_{\rm tr})$ by another difference $h_1(\Theta_{\rm tr}) - h_2(\overline{\Theta}_{\rm tr})$ in a procedure called alignment.[2] The new point $\overline{\Theta}_{\rm tr}$ is defined in such a way that most of its components still coincide with the one of $\Theta_{\rm tr}$; only the ones that belong to parameters from a set S are changed in order to minimize the normalized inner product

$$\frac{\langle h_1(\Theta_{\rm tr}), h_2(\overline{\Theta}_{\rm tr}) \rangle}{\sqrt{\langle h_1(\Theta_{\rm tr}), h_1(\Theta_{\rm tr}) \rangle \langle h_2(\overline{\Theta}_{\rm tr}), h_2(\overline{\Theta}_{\rm tr}) \rangle}} \,. \tag{A7}$$

In principle, S can contain an arbitrary combination of parameters because there is no restriction as to which parameters can be optimized over in (A7). From the motivation, it is natural to select parameters for which conventional differences might occur and as a basic (yet effective) version of S, we choose to include relative time and phase shifts t_0, ϕ_0 (so that (A7) coincides with our definition of the 'overlap'). This also allows a computationally very efficient optimization, exploiting that (A2) can be written as an inverse Fourier transform [126]. Of course, the estimate (A3) has to be changed accordingly, and the details of how to do this are presented in section III. B. of Ref. [28] (the basic structure remains the same, which is why we do not cover it explicitly here).

Appendix B: Additional details of simulations and PE results

In the examples we considered in section V, we find that it is primarily the $\delta\phi$ parameter (for both the parametrizations) that is constrained by the data. We obtain the prior distribution only for the $\delta\tilde{A}$. It can be understood because, in our simulations, we introduce $1\% - 2\%$ relative errors in amplitude. It seems too low for the simulations to pick up variation of that order in the amplitude parameter. Moreover, a constant shift of $\delta\tilde{A}$ is expected to be absorbed in distance posteriors. For the 2G PE analysis, we can use WF-Error parametrizations (12), and (13) which uses only $\delta\tilde{\phi}$ parameter. We choose to keep the $\delta\tilde{A}$ in our parametrizations to account for most general cases of deviation.

In Figure 13, we show the relative amplitude and phase errors between $h_{\rm no-spin}$ and 500 simulated $h_{\rm prec}$ GW strain for the GW150914-like system. The details of the waveform parameters are given in section V C. In PE, we use ten frequency nodal points with logarithmic spacing

[2] Here we assume estimate (A3) to be evaluated in $\Theta_{\rm tr}$, without loss of generality (the discussion also applies when using $\Theta_{\rm bf}$).